Quadratic Coefficients (Three Variables)

(x1 x2 x3) [a11 a12 a13] [x1] = [a11 x1 + a21 x2 + a31 x3 a12 x1 + a22 x2 + a32 x3 a13 x1 + a23 x2 + a33 x3] [x1]
[a21 a22 a23] [x2] (a11 x1 + a21 x2 + a31 x3) x1 + (a12 x1 + a22 x2 + a32 x3) x2 + (a13 x1 + a23 x2 + a33 x3) x3
[a31 a32 a33] [x3]

= a11 x1^2 + a22 x2^2 + a33 x3^2 + (a21+a12) x1 x2 + (a31+a13) x1 x3 + (a32+a23) x2 x3 => Quadratic form

a11 => Coefficient of X1^2
a22 => Coefficient of X2^2
a33 => Coefficient of X3^2

a21+a12 => Coefficient of x1 x2 => a12=a21=1/2 coeff of x1 x2

a13+a31 => Coefficient of x1 x3 => a13=a31=1/2 coeff of x1 x3

a32+a23 => Coefficient of x2 x3 => a32=a23=1/2 coeff of x2 x3

[x1 x2 x3] [a11 a12 a13] => [a11 a12=a21 a13=a31] [coef x1^2 0.5 coef x1x2 0.5 coef x1x3]
[a21 a22 a23] [a12 a22 a23=a32] [0.5 coef x1x2 coef x2^2 0.5 coef x2x3]
[a31 a32 a33] [a13 a23 a33] [0.5 coef x1x3 0.5 coef x2x3 coef x3^2]

(Symmetric matrix - off-diagonal elements are half the corresponding cross-term coefficient)